
Understanding Our Users

Empathy Maps & Design Strategy

Building Inclusive Transit Solutions for Egypt

How user research, empathy mapping, and human-centered design drive feature decisions and create an app that serves diverse users—from daily commuters to elderly travelers, budget-conscious students to international visitors.

Empathy Maps Reveal What Numbers Cannot

Beyond Demographics

Traditional market research tells us **how many users** exist and **where they live**. But empathy maps reveal **why they travel**, **what frustrates them**, **what they hope for**, and **what barriers prevent them** from using digital solutions.

8-12M

Daily transit users in Greater Cairo

The Human Side of Transit

In the Egyptian transit market, users range from **tech-savvy professionals** seeking optimization to **elderly commuters** with accessibility needs. Empathy is not just a design principle—it's **essential for creating an inclusive, usable app** that serves everyone.

4 Personas

Distinct user segments with unique needs

From Insights to Action

Empathy maps transform research into **actionable design decisions**. When we understand that Amira needs real-time reliability, Mohamed needs simplicity, Fatima needs transparency, and Karim needs safety, we can design features that serve all of them.

5 Universal Needs

Reliability, clarity, accessibility, safety, efficiency

Building Inclusive Solutions

Empathy-driven design ensures that no user segment is left behind. By understanding each persona's pain points, motivations, and constraints, we create an app that is **truly valuable** for the diverse population of Greater Cairo.

100%

Target: Serve all user segments effectively

Four Personas, Four Distinct Needs, One App



Amira

Daily Commuter

Age 28

Efficiency & Reliability

Primary Need: Real-time tracking

Market Share: 40%



Mohamed

Elderly Commuter

Age 62

Simplicity & Support

Primary Need: Accessibility

Market Share: 15%



Fatima

Budget-Conscious Student

Age 21

Cost Savings & Transparency

Primary Need: Student discounts

Market Share: 25%



Karim

Casual Visitor

Age 35

Safety & Clarity

Primary Need: English support

Market Share: 20%

Amira's World: 45-60 Minutes Lost to Commuting

A Marketing Manager's Daily Struggle with Fragmented Transit

Amira is a 28-year-old marketing manager commuting from Nasr City to New Cairo daily. She spends **45-60 minutes each way** navigating a fragmented transit system using multiple apps, none of which provide real-time reliability. Her commute is wasting valuable time she could use for work, and she's constantly worried about being late to important meetings.

DAILY COMMUTE

45-60 minutes each way

TRANSIT MODES

Metro + Bus + Microbus

APPS USED

Google Maps, Cairo Metro, Multiple

PRIMARY PAIN

Unpredictable delays & crowding



WHAT SHE SAYS

"I spend too much time figuring out which route to take"

"I wish I could see real-time updates so I'm not surprised by delays"



WHAT SHE THINKS

My commute is wasting valuable time I could use for work

I'm worried about being late to important meetings



WHAT SHE DOES

Checks Google Maps and Cairo Metro app multiple times daily

Switches between different transit modes strategically



WHAT SHE FEELS

Frustrated by unpredictable schedules and crowding

Anxious about being late to work or meetings

Mohamed's Challenge

Profile

Age: 62 years old

Occupation: Retired government employee

Location: Helwan, travels to central Cairo

Tech Savviness: Low (basic phone user)

Frequency: 2-3 trips per week

Journey Duration: 90+ minutes each

Primary Barriers

- Complex app interfaces overwhelm him
- Small text sizes are hard to read
- Doesn't trust digital payment systems
- Prefers human interaction over technology

Says

"These new apps are too complicated for me"

- Buttons are too small to see
- Why so many different apps?
- Prefers talking to people

Thinks

"Technology is moving too fast for me"

- Doesn't trust digital systems
- Feels stupid when confused
- Grandchildren understand better

Does

Relies on traditional methods

- Asks staff for directions
- Writes down route info
- Arrives very early
- Calls family for help

Feels

Anxious, confused, isolated

- Frustrated by complexity
- Dependent on others
- Vulnerable in crowds
- Hopeful about support

Fatma's Dilemma

Every Pound Counts in a Budget

Fatima, 21, is an engineering student commuting daily on a tight budget. Tech-savvy but highly price-sensitive.

WHAT SHE SAYS

"Every pound counts for me" — also looks for student discounts.

WHAT SHE THINKS

Wants to find and compare cheaper routes quickly; worries about hidden costs.

WHAT SHE DOES

Chooses the cheapest transit, walks to save fares, and tracks spendings simply.

WHAT SHE FEELS

Stressed about limited funds, frustrated by opaque pricing, proud when saving money.



25%

Market Share



High Growth

Potential



5-6x

Weekly Commutes



Very High

Tech Savviness

Karim's Uncertainty: Navigating an Unfamiliar City

Karim (Age 35) is a business consultant visiting Cairo for a conference. He's unfamiliar with the local transit system and concerned about language barriers, safety, and making mistakes while exploring the city.



What He Says

"I don't understand the transit system here"

"I want to avoid getting lost in unfamiliar areas"

"Can I pay with my international card?"

"Is this area safe to travel at this time?"



What He Does

Uses Google Maps as primary navigation tool

Asks hotel staff for transit recommendations

Takes taxis/Uber when unsure about public transit

Takes photos of transit maps and routes

Asks other travelers for advice



What He Thinks

Cairo's transit system is different from what he's used to

Should be able to use phone to navigate like in other cities

Worried about making mistakes or being overcharged

Wants to experience the city but stay safe



What He Feels

Uncertain about navigating an unfamiliar system

Anxious about safety and getting lost

Frustrated when apps don't provide needed information

Cautious about trusting unfamiliar payment systems

What All Users Share: Five Universal Needs

1 Reliability

Predictable, consistent service that users can count on daily.

Impact:

Accurate information that supports everyday decisions.

2 Clarity

Transparent, easy-to-understand information to reduce confusion.

Impact:

Clear content helps users decide quickly and confidently.

3 Accessibility

Design that ensures everyone can use the system regardless of ability.

Impact:

Inclusive features that keep all users served.

4 Safety

Users must feel secure both physically and with their data.

Impact:

Safety measures that build trust in the service.

5 Efficiency

Time-saving flows that respect users' schedules.

Impact:

Faster journeys and fewer unnecessary steps.

✓ Design Principle

Use these needs as a lens for every design choice.

Result:

Cohesive features that serve all personas.

Flexible Complexity: One App, Multiple Modes

Designing for both accessibility and power users through intelligent interface adaptation

Simple Mode

For Mohamed & accessibility-focused users

- ✓ **Large, Clear Buttons**
Minimum 48px touch targets for easy tapping
- ✓ **Minimal Options**
Only essential features on each screen
- ✓ **Voice Guidance**
Audio announcements and screen reader support
- ✓ **Step-by-Step Instructions**
Visual and written guidance at each step
- ✓ **Human Support**
Easy access to chat or phone assistance

TARGET USERS

Elderly commuters, users with disabilities, non-technical users, first-time app users

Advanced Mode

For Amira & power users

- ✓ **Real-Time Tracking**
Live vehicle positions and arrival predictions
- ✓ **Multi-Modal Optimization**
Compare routes across all transit modes
- ✓ **Occupancy Information**
See crowding levels before boarding
- ✓ **Analytics & Insights**
Spending trends and commute time analytics
- ✓ **Calendar Integration**
Sync with work calendar for meeting alerts

TARGET USERS

Tech-savvy professionals, frequent commuters, optimization-focused users, power users

Transparent Pricing

Fatima's Insight Shapes Features

Fatima's struggle with opaque pricing reveals a clear need: budget-conscious users require full cost visibility to choose confidently.

THE PROBLEM

Transit apps hide prices, require extra taps to view costs, and rarely compare modes. Users waste time finding the cheapest route.

Our Solution

- Display all costs upfront on route selection
- Show cost breakdown by mode and segment
- Highlight savings and cheaper alternatives
- Integrate student discount eligibility checks
- Provide spending analytics and budget tracking



Cost Comparison Tool

Side-by-side pricing for routes, showing total and per-segment costs with travel time.

Benefit:

Users instantly see the cheapest option



Student Discount Integration

Auto-detect student status and apply eligible discounts at checkout.

Benefit:

Fatima saves on weekly commute costs automatically



Spending Analytics

Weekly/monthly summaries and trend insights to spot savings opportunities.

Benefit:

Track spending and identify savings



Group Discounts & Referrals

Earn discounts via referrals or group commutes and see potential savings.

Benefit:

Community-driven cost reduction

EXPECTED IMPACT

Transparent pricing increases confidence, reduces decision fatigue, and drives adoption among students and budget-conscious commuters.

Real-Time Reliability

Amira's Need for Certainty: Predictable, Trustworthy Information

1

Live Vehicle Tracking

See exactly where buses, metro, and monorail vehicles are in real-time with accurate arrival predictions

2

Service Disruption Alerts

Instant notifications when delays, cancellations, or service changes occur on your planned route

3

Occupancy Information

Know crowding levels before boarding so you can choose less crowded vehicles or alternative routes

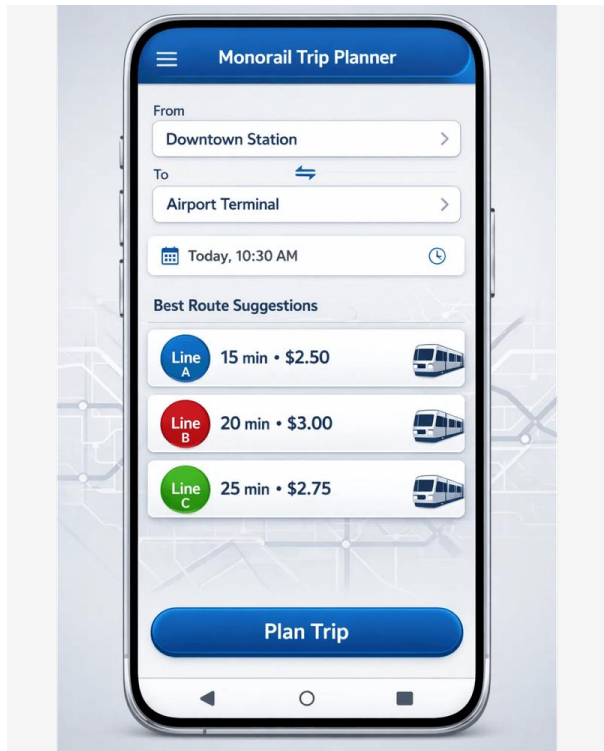
4

Alternative Route Suggestions

When delays occur, the app automatically suggests faster alternative routes to keep you on schedule

WHY THIS MATTERS FOR AMIRA

Amira spends 45-60 minutes commuting daily and is constantly worried about being late to work. Real-time reliability transforms her experience from anxious uncertainty to confident predictability, allowing her to plan her day with confidence.



Accessibility-First Design: Mohamed's Requirements

Universal accessibility standards that ensure no user is left behind, regardless of age, ability, or technical expertise

Visual Accessibility

For users with vision challenges

- ✓ **Large Text**
Minimum 16px body text, 24px+ for headers
- ✓ **High Contrast**
WCAG AAA contrast ratios for readability
- ✓ **Screen Reader Support**
Full semantic HTML for assistive technologies
- ✓ **Text Scaling**
Users can increase text size up to 200%

Interaction Design

For users with motor or cognitive challenges

- ✓ **Large Touch Targets**
Minimum 48px buttons for easy tapping
- ✓ **Voice Input**
Speak destinations and commands
- ✓ **Keyboard Navigation**
Full app functionality without touch
- ✓ **Simplified Navigation**
Minimal steps to complete tasks

Language & Localization

For Arabic-speaking and multilingual users

- ✓ **Arabic-First Design**
Arabic as primary language, English as secondary
- ✓ **Right-to-Left Support**
Full RTL layout for Arabic text
- ✓ **Clear Language**
Simple, everyday Arabic without jargon
- ✓ **Audio Announcements**
Spoken instructions in Arabic and English

Support Systems

For users who need human assistance

- ✓ **Live Chat Support**
Real-time help from trained agents
- ✓ **Phone Support**
Call center for users who prefer voice
- ✓ **Comprehensive FAQ**
Video tutorials and step-by-step guides
- ✓ **Station Staff Training**
Agents trained to assist elderly users

Multilingual & Culturally Sensitive Design

Meeting international visitors and diverse users where they are

CORE PRINCIPLE

Arabic-first interface with seamless English support. Every feature, instruction, and safety message is available in both languages from day one.

Language & Communication

INTERFACE

- ✓ Arabic as primary language with native speakers designing UI copy
- ✓ English as secondary language for international users
- ✓ Easy language toggle in settings (no page reload required)

SUPPORT

- ✓ Arabic-speaking support team available via chat and phone
- ✓ English-speaking support for international users

Payment Methods

- ✓ Local methods: Vodafone Cash, Etisalat Cash, Orange Money
- ✓ International: Visa, Mastercard, Apple Pay, Google Pay
- ✓ Clear currency display and exchange rate transparency

Safety & Cultural Context

SAFETY INFORMATION

- ✓ Real-time safety alerts for each station and route
- ✓ Women-only carriage information and schedules
- ✓ Peak hour crowding warnings and alternative routes
- ✓ Emergency contact integration and SOS features

CULTURAL SENSITIVITY

- ✓ Prayer time notifications and mosque location information
- ✓ Ramadan-specific schedules and adjusted transit times
- ✓ Halal food and prayer facility guides at major stations
- ✓ Respectful imagery and culturally appropriate messaging

IMPACT

Visitors like Karim feel welcomed and confident navigating Cairo. Local users appreciate a system designed for their culture and values. Trust increases, adoption accelerates.

Social & Community Features

Fatima's Insight: Building Loyalty Through Community

CORE INSIGHT			
Fatima is not just looking for cheap transit—she's looking for belonging and shared experiences. Social features transform the app from a transactional tool into a community platform where users support each other in saving money and sharing commute experiences.			
Ride-Sharing & Matching Connect users with similar routes and schedules	Student Community & Discounts Exclusive features for student users like Fatima	Referral & Loyalty Program Reward users for growing the community	Community Engagement Build belonging and shared purpose
✓ Commute Matching Find users traveling the same route at the same time ✓ Shared Taxi Pooling Split taxi costs with other commuters ✓ Safety Verification User ratings and verified identity badges	✓ Student Community Hub Discussion forums and tips for budget commuting ✓ Group Discounts Bulk discounts when 5+ students book together ✓ University Partnerships Exclusive deals with major universities	✓ Referral Rewards Earn transit credits for each friend who joins ✓ Loyalty Tiers Bronze, Silver, Gold status with escalating benefits ✓ Milestone Rewards Bonuses at 10, 50, 100 trips completed	✓ User Reviews & Tips Share route recommendations and transit tips ✓ Leaderboards Monthly rankings for most eco-friendly commuters ✓ Social Challenges "Green Commute Week" with rewards for participation
IMPACT Users save 30-50% on shared taxi costs while building friendships and support networks	IMPACT Students get 25-40% discounts while universities gain student engagement and recruitment advantages	IMPACT Viral growth through word-of-mouth while users accumulate free transit credits through loyalty	IMPACT Users feel part of a movement improving Cairo's mobility while supporting each other financially

Personalization & Adaptation

Smart User Experience

CORE CONCEPT

AI learns each user's preferences, behavior patterns, and needs, then automatically adapts the interface, notifications, and recommendations. One app, four different experiences—each perfectly tuned to individual personas.

Interface Adaptation

Complexity that matches user skill

✓ **Auto-Detect User Type**

Identifies whether user prefers simple or advanced mode

✓ **Progressive Disclosure**

Show advanced features only when user is ready

✓ **Font Size Adjustment**

Automatically increase text for elderly users

✓ **Language Preference**

Remember Arabic/English choice and usage patterns

Example: Mohamed gets simple mode; Amira gets advanced features

Smart Recommendations

Suggestions tailored to behavior

✓ **Favorite Routes**

Learn and suggest frequently used routes

✓ **Cost Optimization**

Suggest cheapest routes for budget-conscious users

✓ **Time Optimization**

Prioritize fastest routes for time-sensitive users

✓ **Comfort Preferences**

Suggest less crowded vehicles and times

Example: Fatima sees cheapest routes first; Amira sees fastest routes

Contextual Notifications

Alerts that respect user preferences

✓ **Frequency Control**

Learn how often user wants notifications

✓ **Timing Optimization**

Send alerts at times user is most likely to act

✓ **Relevance Filtering**

Only show alerts that matter to this user

✓ **Quiet Hours Respect**

No notifications during user's sleep time

Example: Karim gets safety alerts; Fatima gets price drop notifications

Multiple Support Channels

Meeting Users Where They Are

Diverse support options that respect different user preferences and communication styles



In-App Chat

Real-time messaging with trained support agents available 24/7

- Instant responses to urgent questions
- Conversation history saved in app

Best for: Tech-savvy users, immediate assistance needs



Phone Support

Call center with Arabic-speaking agents for detailed assistance

- Live agent support during business hours
- Callback option to avoid long waits

Best for: Elderly users, complex issues, preference for voice



WhatsApp Support

Messaging through WhatsApp for users who prefer this platform

- Send screenshots and photos for help
- Asynchronous communication (no need to wait)

Best for: Users familiar with WhatsApp, non-urgent questions

DESIGN PRINCIPLE: NO USER LEFT BEHIND

Offer multiple channels (text, voice, video, community) so every user can access help in their preferred way — increasing trust and satisfaction.



FAQ & Knowledge Base

Comprehensive self-service resources for independent learners

- Searchable knowledge base

Best for: Self-sufficient users, learning at own pace



Community Forum

Peer-to-peer support where users help each other

- User community sharing tips and tricks

Best for: Community-oriented users, building loyalty



Email Support

Formal support channel for detailed inquiries and feedback

- Ticket tracking system

Best for: Formal complaints, detailed documentation needs

Bringing Empathy to Life

Phased Implementation Roadmap

A 12-month development journey from core functionality to advanced personalization

1

Foundation

Months 1-3

CORE FEATURES

- ✓ Trip planning & route search
- ✓ Real-time tracking integration
- ✓ Ticket booking & QR codes
- ✓ Payment processing setup

ACCESSIBILITY BASICS

- ✓ Arabic-first interface design
- ✓ Large text & high contrast
- ✓ Screen reader compatibility
- ✓ Basic support channels

2

Personalization

Months 4-6

USER ADAPTATION

- ✓ Simple vs. advanced mode toggle
- ✓ Favorite routes & saved preferences
- ✓ Transparent pricing features
- ✓ Student discount integration

COMMUNITY FEATURES

- ✓ Referral program launch
- ✓ User reviews & ratings
- ✓ Community forum setup
- ✓ Loyalty tier system

IMPLEMENTATION PHILOSOPHY

Each phase builds on empathy insights: Foundation establishes accessibility for all users; Personalization adapts to individual needs; Intelligence anticipates user preferences; Scale & Optimize extends impact across Greater Cairo. Every feature is evaluated against persona needs and accessibility standards.

3

Intelligence

Months 7-9

AI & ANALYTICS

- ✓ Smart recommendations engine
- ✓ Spending analytics & insights
- ✓ Predictive delay alerts
- ✓ Occupancy predictions

ADVANCED SUPPORT

- ✓ WhatsApp support integration
- ✓ AI chatbot for common questions
- ✓ Video tutorial library

4

Scale & Optimize

Months 10-12

ADVANCED FEATURES

- ✓ Ride-sharing matching system
- ✓ Calendar integration & alerts
- ✓ Multi-modal journey planning

OPTIMIZATION & GROWTH

- ✓ Performance optimization
- ✓ Marketing campaign launch
- ✓ Partnership integrations
- ✓ Expansion to other cities

Measuring Empathy

KPIs That Matter

Success metrics that go beyond downloads to measure real user value delivery and persona satisfaction

Engagement Metrics	
Daily Active Users (DAU)	Target: 45%
Trip Bookings per User	Target: 8-12 trips/mo
Feature Adoption Rate	Target: 60%
Session Duration	Target: 5-8 min

Satisfaction Metrics	
Net Promoter Score (NPS)	Target: 50+
App Store Rating	Target: 4.5+
Customer Satisfaction (CSAT)	Target: 80%
Support Ticket Resolution	Target: 90%

Business Metrics	
User Acquisition Cost (UAC)	Target: EGP 5-10
Lifetime Value (LTV)	Target: LTV:UAC 5:1
Churn Rate	Target: <5%
Referral Rate	Target: 30%

Persona-Specific Metrics	
Amira (Commuter) Retention	Target: 85%
Mohamed (Elderly) Accessibility	Target: 70%
Fatima (Student) Cost Savings	Target: 25-30%
Karim (Visitor) Safety Score	Target: 4.3+

MEASUREMENT PHILOSOPHY

We measure success not just by downloads, but by real user value—time saved for commuters, money saved for students, confidence for elderly users, and safety for visitors. Persona-specific metrics ensure empathy-driven design delivers tangible benefits.

Empathy as Competitive Advantage

How designing with empathy for all personas creates an inclusive, valuable app that improves mobility across Greater Cairo



Market Differentiation

While competitors focus on features, we focus on people.

- Arabic-first design that respects local culture
- Accessibility for all ages and abilities
- Multiple support channels meeting diverse preferences
- First mover advantage in empathy-driven transit



User Impact

Real benefits for every persona, every day.

- Amira saves 30+ minutes weekly through optimization
- Mohamed feels confident and supported
- Fatima saves 25-30% on transit costs
- Karim feels safe and welcomed in Cairo



Business Impact

Empathy drives sustainable growth and loyalty.

- Higher retention through genuine user value
- Viral growth via word-of-mouth referrals
- Premium positioning justifies higher margins
- Defensible market position through loyalty

The Empathy Advantage

In a market where competitors focus on speed and features, we focus on understanding. By deeply empathizing with Amira's time pressures, Mohamed's accessibility needs, Fatima's budget constraints, and Karim's safety concerns, we create an app that doesn't just move people—it improves their lives. This human-centered approach creates genuine loyalty, drives organic growth, and establishes a defensible competitive position in the Cairo transit market.

The Transit App: Not just how you move. How you belong.

Next Steps: From Empathy to Action

Actionable guidance for every team to implement empathy-driven design decisions



Design Team

Create interfaces that adapt to user needs

- Implement simple vs. advanced mode toggle in wireframes
- Design for 16px minimum body text and 24px+ headers
- Create Arabic-first mockups with RTL layout support
- Design 48px+ touch targets for accessibility
- Establish design system with persona-specific variations



Marketing Team

Communicate persona-specific value propositions

- Create persona-specific marketing campaigns for each segment
- Highlight accessibility and Arabic-first design in messaging
- Develop student discount and referral program promotions
- Partner with universities, elderly centers, and visitor bureaus
- Create testimonial videos featuring real personas using the



Product Team

Prioritize features that deliver persona value

- Map each feature to at least one persona's core need
- Prioritize transparent pricing and cost comparison
- Plan real-time tracking and service alerts for Phase 1
- Define student discount and referral program mechanics
- Create accessibility requirements checklist for all features



Support Team

Provide help in the way users prefer

- Establish 24/7 in-app chat support with Arabic agents
- Set up phone support line for elderly and non-tech users
- Create comprehensive FAQ and video tutorial library
- Integrate WhatsApp support channel for accessibility
- Train support staff on persona-specific communication styles



Development Team

Build robust, accessible, and performant systems

- Implement semantic HTML for screen reader compatibility
- Build Arabic language support with RTL CSS framework
- Create feature flags for simple/advanced mode switching
- Integrate real-time tracking APIs and live data feeds
- Set up analytics to track persona-specific metrics

IMPLEMENTATION TIMELINE

Phase 1 (Months 1-3): Foundation + Accessibility. Phase 2 (Months 4-6): Personalization + Community. Phase 3 (Months 7-9): Intelligence + Advanced Support. Phase 4 (Months 10-12): Scale + Optimization. Success depends on every team embracing empathy as a core principle, not an afterthought.